

Tamlin Love

Barcelona, Spain

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RESEARCH INTERESTS

I am a young researcher from South Africa, now working as a PhD student at the Institut de Robotica i Informàtica Industrial (IRI) (under CSIC and UPC) in Barcelona, working on the TRAIL MSCA-DN project. I am passionate about the possibilities for humans and robots to communicate, collaborate and learn from each other. Right now, my primary interests lie in explainable robotics, where robots must explain the reasons for their decisions and beliefs. I am interested not only in how these explanations are constructed, but also in how they fit into broader human-robot interactions. More generally, I am interested in other areas of human-robot interaction, including personalisation and human-in-the-loop learning. I am also an active member and promoter of the African robotics community, and have organised workshops and conferences with the aim of growing robotics on the African continent.

EDUCATION

Universitat Politècnica de Catalunya <i>PhD in Automatic Control, Robotics and Computer Vision</i>	Barcelona, Spain 2023 – Present
University of the Witwatersrand <i>Master of Science in Computer Science (with distinction)</i>	Johannesburg, South Africa 2020 – 2021
University of the Witwatersrand <i>Bachelor of Science Honours in Computer Science (with distinction)</i>	Johannesburg, South Africa 2019
University of the Witwatersrand <i>Bachelor of Science in Computer Science (with distinction)</i>	Johannesburg, South Africa 2016 – 2018

Member of the [Perception and Manipulation](#) group at the [Institut de Robòtica i Informàtica Industrial](#) from 2023 - present.
Member of the [Robotics, Autonomous Intelligence and Learning](#) (RAIL) from 2020 - 2023.

RECENT PUBLICATIONS AND PRE-PRINTS

For a complete list, please follow [this link](#).

<u>HEXAR: a Hierarchical Explainability Architecture for Robots</u> <i>Tamlin Love, Ferran Gebellí, Pradip Pramanick, Antonio Andriella, Guillem Alenyà, Anaís Garrell, Raquel Ros, Silvia Rossi</i> 2026 IEEE International Conference on Robotics & Automation (Best Paper in HRI)	2026
<u>Causal Explanations for Robot Decisions and Beliefs</u> <i>Tamlin Love, Antonio Andriella, Guillem Alenyà</i> 2026 ACM/IEEE International Conference on Human-Robot Interaction (Pioneers Workshop)	2026
<u>Temporal Counterfactual Explanations of Behaviour Tree Decisions</u> <i>Tamlin Love, Antonio Andriella, Guillem Alenyà</i> Under review	2025
<u>Measuring User Understanding in Explainable Human-Robot Interaction: A systematic Review</u> <i>Ferran Gebellí, Pradip Pramanick, Tamlin Love, Raquel Ros, Anaís Garrell, Silvia Rossi, Antonio Andriella, Guillem Alenyà</i> Under review	2025
<u>What Would I Do If...? Promoting Understanding in HRI through Real-Time Explanations in the Wild</u> <i>Tamlin Love, Antonio Andriella, Guillem Alenyà</i> 2024 IEEE International Conference on Robot and Human Communication	2024

WORK EXPERIENCE

Researcher <i>Institut de Robòtica i Informàtica Industrial</i>	2023 – Present Barcelona, Spain
- Working on the TRAIL: TRAnsparent InterpretabLe robots project (project code: HORIZON-MSCA-DN-101072488), a Marie Skłodowska-Curie Doctoral Network project funded by the European Union. - Involves research on explainability, causality, human-robot interaction, and robot perception and decision-making. - Includes extensive use of Python and ROS.	

Sessional Lecturer

University of the Witwatersrand

2022 – 2023

Johannesburg, South Africa

- Managed and taught the 3rd year Software Design courses and co-taught the 1st year Basic Computer Organisation course for the Computer Science undergraduate degree
- Set, oversaw and marked practical and theoretical assessments for the above courses

Backend Developer

BodyBack

2021 – 2022

Johannesburg, South Africa

- Created and maintained the REST backend framework for the BodyBack mobile application

Tutor

University of the Witwatersrand

2018 – 2021

Johannesburg, South Africa

- Tutored a number of computer science and applied mathematics courses for undergraduate and honours students

COMMUNITY BUILDING

Founder and Academic Partnerships Chair

Akili African Robotics Community

2026 – Present

- I am a founding member of the organising committee for the [Akili African Robotics Community \(AARC\)](#), a network of roboticists focused on promoting accessibility, community and education in the African robotics space. Through this organisation, I have co-organised several workshops, satellite events, and a monthly webinar series.

RESEARCH STAYS

Academic Secondment

RAIL Lab, University of the Witwatersrand

April 2026 – May 2026

Johannesburg, South Africa

- This stay at the [RAIL laboratory](#), under the supervision of Benjamin Rosman, formed part of the EMBRACE project (funded by the CSIC I-COOP). During the stay, I implemented a human-robot collaborative sandwich making task on a UR5 robot arm, using behaviour trees to execute high-level decision-making under uncertainty.

Academic Secondment

PRISCA Lab, Università degli Studi di Napoli

June 2025 – August 2025

Naples, Italy

- This stay at the [PRISCA laboratory](#), under the supervision of Silvia Rossi as part of the TRAIL doctoral network, culminated in the development of the HEXAR framework for hierarchical explainability, implemented on the PAL TIAGo robot, in collaboration with partners from the laboratory and PAL robotics.

Industrial Secondment

ZAL Center of Applied Aeronautical Research

March 2025

Hamburg, Germany

- This stay at [ZAL](#), under the supervision of Martin Gromniak as part of the TRAIL doctoral network, focused on industry exposure. I spent the research stay evaluating the performance of different object detection models on a dataset of cabin damages.

PROGRAMMING EXPERIENCE

Languages : Python, Java, C, C++, Matlab

I have experience working with several robot platforms (ARI, Kuri, OTE RealAnt, TIAGo, TurtleBot, UR5) and with the robotics operating system (ROS1 and ROS2).

OTHER EXPERIENCE

Organisation

- Sponsorship chair of the HRI Pioneers Workshop in at HRI in the USA, 2027 ([Website](#))
- “Robotic Systems Designed for Underrepresented Communities” forum at ICRA 2026 ([Website](#))
- General chair of the Transparent and Interpretable Robots (TRAIL) Conference in Italy, 2026 ([Website](#))
- “Workshop on Robotics and Automation in Africa” at the Deep Learning Indaba ([Nigeria 2026](#); [Rwanda 2025](#); [Senegal 2024](#); [Ghana 2023](#))
- Explainable Human-Robot Interaction Special Session @ RO-MAN, 2025

- IEEE “ICRA@40 Africa” satellite conference in South Africa, 2024 ([Website](#))
- “From Specialists to Generalists: The World of Generalised RL Agents” workshop at the Deep Learning Indaba in Tunisia, 2022 ([Website](#))

Conferences and Summer Schools

Deep Learning Indaba (Tunis 2022, [Dakar 2024](#)); EASE Fall School ([Bremen 2023](#)); ECAI ([Santiago de Compostela 2024](#)); HRI ([Boulder 2024](#) virtual, [Edinburgh 2026](#)); ICANN ([Lugano 2024](#)); ICRA ([Vienna 2026](#)); RO-MAN ([Pasadena 2024](#)); RSE ([Barcelona 2026](#))

Reviewing

AAMAS 2024; HRI 2024, 2026; ICRA 2023, 2026; ICSR 2023, 2025; IEEE RA-L 2025; IROS 2024; NeurIPS 2022; ROBOT 2026; RO-MAN 2024–2026

Invited Talks

University of Pretoria 2022; [IndabaX South Africa 2022](#); Saarland University 2024; Universitat Autònoma de Barcelona 2025;

Languages

- English (home language)
- Italian (conversational)
- Spanish (conversational)
- Catalan (basic)

AWARDS AND HONOURS

Fellowships

- Selected as an HRI Pioneer at the HRI conference in Edinburgh, 2026

Conference Awards

- Best paper in HRI award at ICRA in Vienna, 2026
- Poster award at the Deep Learning Indaba in Tunis, 2022

Academic Awards

- Post-graduate merit award scholarship at the University of the Witwatersrand, 2019–2022
- Liberty Life Gold Medal for outstanding performance in Computer Science, 2018
- Prize for Top Student in Computational & Applied Mathematics at the University of the Witwatersrand, 2018
- Herbert Le May Prize for the best 3rd year science student majoring in Applied Mathematics at the University of the Witwatersrand, 2018
- 2nd place in the SAMMC competition for mathematical modelling, 2018
- Psybergate Computer Science prize at the University of the Witwatersrand, 2017